Koa Shen

ROBOTICS ENGINEERING | SUMMER 2027

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PROFESSIONAL SUMMARY

Mechanical engineering student focused on robotics, mechanisms, and motion control. Experience spans kinematic simulation, CAD, custom actuator design, embedded stepper and encoder control, precision fabrication, and hands-on hardware validation.

EDUCATION

University of California, Santa Barbara | B.S. Mechanical Engineering
Santa Barbara, CA | Expected June 2028 | Major GPA: 3.97 | Cumulative GPA: 3.94
Dean's Honors every quarter | Relevant coursework: Dynamics, Mechanics of Materials, Circuits, MATLAB, Dynamical Systems

ROBOTICS AND ENGINEERING EXPERIENCE

Mechanical Engineering Intern | SLAC National Accelerator Laboratory, LCLS
Summer 2026–Present | Menlo Park, CA; extended part-time remote

- Built a Python/Drake digital twin for a 30-DOF, 305-body beamline motion assembly with live collision checking; millisecond checks exposed at least five physical pinch points.
- Developed 15 reusable motion-stage models; the framework is being adopted across three LCLS assemblies.
- Designed optics mounts and alignment features in Siemens Solid Edge; applied GD&T and Teamcenter PDM documentation for manufacture and inspection.

Fabrication Team Lead | Gaucho Racing, Formula SAE Electric, UC Santa Barbara
Fall 2024–Present | Santa Barbara, CA

- Led a 15-person team through driver-safety-system design, fabrication, and installation; GR26 passed Formula SAE technical inspection.
- CNC-machined 33 steel and aluminum suspension parts, holding critical bearing interfaces within 0.0005 in.
- Contributed to steering-wheel integration of buttons, potentiometers, a display, and quick-release hardware; FEA-guided iterative design reduced mass from 5 lb to 1.7 lb.

Maintenance Technician | Exploratorium
Summer 2025 | San Francisco, CA

- Diagnosed and repaired mechanisms across 100+ public exhibits; fabricated replacement parts using manual machining, welding, and shop tools.
- Redesigned encoder mounts for reliability and serviceability and documented CAD and remanufacturing instructions for future technicians.

ROBOTICS PROJECTS

Closed-Loop Cycloidal Actuator | Personal Project

- Designed and fabricated a 15:1 cycloidal reducer within a NEMA 17 motor's 42 mm square footprint; integrated a TMC2209 stepper driver and AS5600 motor-shaft encoder.
- Developed PlatformIO C++ firmware and bench-validated closed-loop position correction and missed-step detection on a single-joint setup.

TECHNICAL SKILLS

Robotics and Controls: Kinematics, Drake, stepper motors, TMC2209, AS5600 encoders, closed-loop position correction, embedded C/C++
Mechanical Design: SolidWorks, Solid Edge, GD&T, FEA, mechanisms, CAD, Teamcenter PDM
Fabrication and Software: CNC/manual machining, 3D printing, welding, Python, MATLAB, PlatformIO, Git